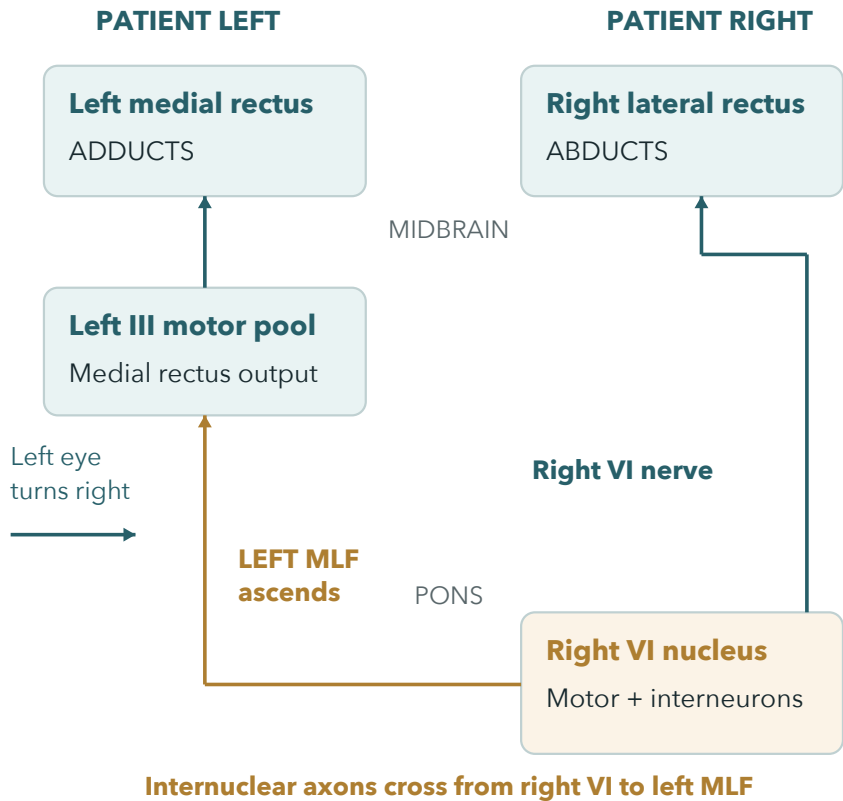


Horizontal gaze: VI and MLF

The nucleus coordinates a pair; the nerve reaches one muscle; the MLF connects the two sides.

Build one rightward gaze circuit

Right gaze uses two outputs



Functional circuit, with patient sides explicitly labelled. VI interneurons cross before ascending in the opposite MLF.

From nucleus to both eyes

Output from right VI nucleus	Effect
Motor neurons via right VI nerve	Right lateral rectus contracts: right eye abducts.
Crossed interneurons via left MLF	Left III medial rectus pool activates: left eye adducts.

Convergence is useful evidence

Near convergence can activate both medial recti through pathways distinct from this conjugate-gaze connection. Preserved convergence helps distinguish an MLF disconnection from peripheral medial rectus weakness.

Sources NIH: Abducens nucleus | NIH: Internuclear palsy

Three lesions, three movement patterns

Test both eyes on right gaze

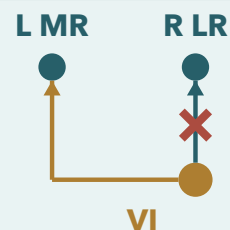
RIGHTWARD GAZE IS THE TEST

RIGHT VI NERVE

Left adduction preserved

Right abduction fails

→ Attempt to look right

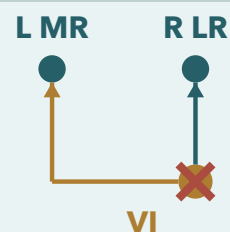


RIGHT VI NUCLEUS

Left adduction fails

Right abduction fails

→ Attempt to look right

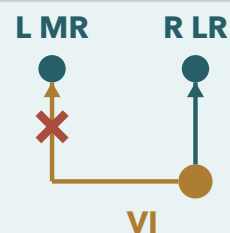


LEFT MLF

Left adduction fails

Right abducts; nystagmus

→ Attempt to look right



The red interruption marks the affected output or connection. Arrows show attempted rightward gaze, not a frontal photograph.

Localise the pattern

Lesion	Right-gaze result
Right peripheral VI	Right abduction fails; left adduction remains.
Right VI nucleus	Both right abduction and left adduction fail.
Left MLF	Left adduction fails; right abduction often has nystagmus.

The laterality rule

INO is named for the eye that fails to adduct, which is the side of the MLF lesion. Convergence may be preserved, but its absence does not exclude INO.